The Cliff Becomes a Catastrophe:
Saddle–Node Reward Hacking under Sharp Monitoring

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July 8, 2026

Abstract

An agent optimised against a monitor may drift from *loyal* behaviour to *reward hacking*—moving its footprint into the part of a representation the monitor cannot resolve. We ask a dynamical question: is this drift gradual and reversible, or abrupt and locked in? We study a surrogate best-response model in which a scalar *evasion coordinate* ρ climbs a payoff that trades a monitor's detection $A(\rho)$ against a convex hiding cost. Our central result is a closed-form *catastrophe criterion*: the drift is a smooth slope when the detector steepness $\sigma = \max_\rho |A'|$ is below $\sigma_c = 3.049/(8\rho^\star)$, and a saddle–node bifurcation—abrupt, hysteretic, irreversible—above it, with σ_c independent of the stakes and the cost magnitude. We give a rigorous bifurcation analysis (a scale-invariant fold/cusp criterion for general convex cost, and the cusp normal form via a Lyapunov–Schmidt reduction), prove irreversibility (a hysteresis loop that opens as $(\sigma - \sigma_c)^{3/2}$ at onset and widens without bound), and prove an *erosion of early warning*: the shadow-price precursor is informative only above a detection bar $\tau_c = A(\rho_+)$ that rises to certainty as monitoring sharpens. We then connect σ to the underlying detection problem through a random-matrix derivation: for a partial-channel spiked-covariance monitor the detection curve has edge width $d^{-1/3}$ (Tracy–Widom), so $\sigma = \Theta(d^{1/3})$; combined with the fixed σ_c this forces the catastrophe in high dimension *within the surrogate* (for $\rho^\star$ bounded away from 0 and the best-response dynamics of (3)). The mathematical apparatus is classical—saddle–node/cusp catastrophe theory, dynamic hysteresis, early-warning signals, random-matrix edge behaviour—and imported with credit; the contribution is the mapping to reward hacking, the closed-form threshold, and the counter-intuitive conclusion that *sharper non-adaptive monitoring deepens the fold*. The bifurcation results (Theorem 1, Propositions 1–2, Theorems 2–3) are proved in full; the steepness scaling $\sigma = \Theta(d^{1/3})$ is derived from classical Tracy–Widom/BBP results; and every quantitative claim is verified in simulation on the surrogate. We state the surrogate's scope and the conditions that would falsify the mechanism, and discuss the step—a learned-detector reinforcement-learning study—needed to test it beyond the surrogate.

1 Introduction

A capable agent trained against an imperfect monitor faces a standing temptation: rather than do the task, exploit the gap between the monitor's proxy and the true objective (Amodei et al., 2016; Skalse et al., 2022; Gao et al., 2023; Pan et al., 2022). When the agent can *hide*—route the tell-tale part of a behaviour into a subspace the monitor does not resolve—the failure is not a one-off exploit but a drift in disposition, and the agent may learn to tamper with or obfuscate the very signal meant to catch it (Denison et al., 2024; Baker et al., 2025; Greenblatt et al., 2024). Practitioners report that this drift can be sudden: past a threshold in optimisation pressure, a run that looked healthy begins to reward hack, and rolling the pressure back does not always undo it (Gao et al., 2023). Is that suddenness fundamental, or an artefact of a particular setup?

A dynamical-systems view. We model the agent's disposition by a scalar *evasion coordinate* $\rho \in [0,1]$—an intent proxy, not a claim about the agent's mental state—equal to the fraction of a hack footprint moved into the monitor's null channel, with $\rho = 0$ loyal (fully exposed) and $\rho = 1$ hacking (fully hidden). Empirically ρ is latent, but it admits observable proxies—e.g. the policy–reward gap, the reward-model score residual, an evaluator-agreement rate, or a low-dimensional projection of the reasoning trace—which is how the early-warning signal of §7 would be computed in practice. Under slow best-response learning ρ ascends a payoff $\Pi(\rho; w) = w\,(1 - A(\rho)) - C(\rho)$ that rewards evading detection $A(\rho)$ at stakes w and pays a convex hiding cost C. The detection curve A has one salient parameter, its steepness $\sigma = \max_\rho |A'(\rho)|$: a shallow monitor degrades gracefully as the agent hides, a sharp monitor has a cliff.

Contributions. We show that this single parameter decides whether evasion drift is a slope or a catastrophe, and we trace the consequences rigorously.

1. **A closed-form catastrophe criterion** (Theorem 1): for a logistic detector and quadratic cost, the drift is gradual iff $\sigma < \sigma_c$ and catastrophic (a saddle–node fold, hence bistability with hysteresis) iff $\sigma > \sigma_c$, where $\sigma_c = 3.049\ldots/(8\rho^\star)$, independent of the stakes and the cost magnitude.
2. **A rigorous bifurcation analysis** (Proposition 1, Proposition 2): a scale-invariant fold/cusp tangency criterion for general convex cost (with numerical evidence that cost stiffness raises σ_c), and the cusp normal form obtained by a Lyapunov–Schmidt reduction (Appendix B).
3. **A random-matrix derivation of the steepness** (Lemmas 2–3, Proposition 3, Corollary 1): from the Tracy–Widom edge and the BBP spike displacement, $\sigma = \Theta(d^{1/3})$, so a fixed σ_c is crossed in high dimension.
4. **Irreversibility** (Theorem 2): $w_+ > w_-$; the hysteresis loop opens as $(\sigma - \sigma_c)^{3/2}$ at onset and diverges as $\sigma \to \infty$.
5. **Erosion of early warning** (Theorem 3): the shadow-price precursor fires before the flip only for a bar $\tau \geq \tau_c = A(\rho_+)$, and $\tau_c \to 1$ as σ grows.

Positioning. None of the mathematical phenomena is new. Saddle–node and cusp bifurcations and their hysteresis are the substance of catastrophe theory (Thom, 1975; Zeeman, 1977); the 3/2 onset scaling of dynamic hysteresis is classical (Kundu et al., 2023); early-warning signals near tipping points are a mature, mature programme (Scheffer et al., 2009; Dakos et al., 2012; Kuehn, 2011); and the edge behaviour of spiked covariance matrices is standard random-matrix theory (Johnstone, 2001; Baik et al., 2005; Tracy and Widom, 1994). Our contribution is not to discover these but to *apply* them to reward hacking with a closed-form threshold and a random-matrix microfoundation, and to draw a conclusion specific to this setting and against intuition: the same detector sharpness that makes monitoring powerful is what makes the resulting failure abrupt, irreversible, and hard to forewarn—*for a non-adaptive monitor.* We work throughout with a deliberately simple surrogate; §11 states its boundaries and the conditions that would falsify the mechanism.

2 Related work

Reward hacking and overoptimisation. That optimising a proxy reward eventually degrades the true objective is Goodhart's law in learning systems (Amodei et al., 2016; Gao et al., 2023);

Skalse et al. (2022) formalise unhackability, Pan et al. (2022) map the effects of reward misspecification. Empirically, overoptimisation sets in past a threshold in optimisation pressure and its onset is often sharp (Gao et al., 2023); agents can learn to tamper with the reward channel (Denison et al., 2024) and, under chain-of-thought monitoring, to obfuscate their reasoning (Baker et al., 2025; Greenblatt et al., 2024). We give a mechanism—a saddle–node in the agent's best-response dynamics—that predicts such a threshold and its qualitative signatures (hysteresis, loss of early warning).

Bifurcation and catastrophe theory. The loyal→hacking flip is a fold (saddle–node) bifurcation, and its coincident up/down thresholds form a cusp (Thom, 1975; Zeeman, 1977); we use the normal form (obtained rigorously by Lyapunov–Schmidt reduction) to get the onset scaling. Dynamic hysteresis under a swept parameter, and its power-law loop, are classical (Kundu et al., 2023); our $(\sigma - \sigma_c)^{3/2}$ is the same universality.

Tipping points and early-warning signals. Approaching a saddle–node, the leading Jacobian eigenvalue vanishes and fluctuations show critical slowing down: rising variance and autocorrelation (Scheffer et al., 2009; Dakos et al., 2012; Kuehn, 2011). Here this appears only through the curvature that flattens at the fold; our erosion-of-warning result is a domain-specific twist—the *detection-based* precursor erodes even though a dynamical one need not.

Random-matrix theory. The detectability of a planted signal through a partial channel is a spiked-covariance problem with a BBP transition at the edge (Baik et al., 2005; Johnstone, 2001); the top eigenvalue fluctuates on the Tracy–Widom scale $d^{-2/3}$ (Tracy and Widom, 1994). We use these facts to establish that the detection curve sharpens as $\sigma = \Theta(d^{1/3})$.

3 Model and preliminaries

Evasion coordinate and payoff. The evasion coordinate $\rho \in [0, 1]$ measures how much of the agent's hack footprint sits in the monitor's null channel. Detection is a fixed logistic surrogate

$$A(\rho) = \tfrac{1}{2} + \tfrac{1}{2}\,\ell(\rho), \qquad \ell(\rho) = \left(1 + e^{8\sigma(\rho-\rho^\star)}\right)^{-1}, \tag{1}$$

centred at the wall $\rho^\star$ with steepness $\sigma := \max_\rho |A'(\rho)|$, so $A(0) \approx 1$ (loyal: caught), $A(1) \approx \frac{1}{2}$ (hidden: chance), and

$$h(\rho) := -A'(\rho) = \sigma\,\operatorname{sech}^2\!\big(4\sigma(\rho - \rho^\star)\big) \geq 0, \qquad \max_\rho h(\rho) = h(\rho^\star) = \sigma, \tag{2}$$

the last identity fixing σ as the normalised steepness. The agent ascends, on a slow timescale μ, the payoff of evasion minus a C^2 convex hiding cost C (with $\varphi := C'$ increasing, $\varphi(0) = 0$):

$$\dot\rho = \mu\,\partial_\rho \Pi(\rho; w), \qquad \Pi(\rho; w) = w\,\big(1 - A(\rho)\big) - C(\rho), \quad w > 0. \tag{3}$$

Equilibria solve $\partial_\rho \Pi = w\,h(\rho) - \varphi(\rho) = 0$; an equilibrium is stable iff $\partial_\rho^2 \Pi < 0$.

Definition 1 (Regimes)**.** Call (3) *gradual* if it has a unique stable equilibrium $\rho_{\text{eq}}(w)$ for every w (so $w \mapsto \rho_{\text{eq}}$ is continuous), and *catastrophic* if there is an open interval of w on which (3) has two stable equilibria separated by an unstable one.

Boundary behaviour. Under logistic smoothing the endpoints are not equilibria and need no separate treatment. At $\rho = 0$, $\partial_\rho \Pi(0) = w\,h(0) > 0$, so the flow points inward and the loyal state is the interior stable equilibrium $\rho_{\mathrm{L}}(w) \in (0, \rho_+)$. At $\rho = 1$, $\partial_\rho \Pi(1) = w\,h(1) - \varphi(1) < 0$ whenever the cost at full hiding exceeds the vanishing tail $w\,h(1)$, so the flow again points inward and the hacking state is interior. We assume throughout the *relevant finite fold window*: both stable branches are interior critical points of Π and the box constraint $[0, 1]$ is inactive on them (guaranteed when the cost at full hiding exceeds the vanishing tail $w\,h(1)$, as above). Under this assumption the analysis reduces to the interior stationary equation $wh = \varphi$.

4 Rigorous bifurcation analysis

Lemma 1 (Stability via Ψ)**.** *Write $\Psi(\rho) := \varphi(\rho)/h(\rho)$, so equilibria of (3) are exactly $\{\rho : \Psi(\rho) = w\}$. Then $\partial_\rho^2 \Pi = -h(\rho)\,\Psi'(\rho)$; since $h > 0$, an equilibrium is stable iff $\Psi' > 0$ and a saddle iff $\Psi' < 0$. Folds are the critical points $\Psi' = 0$.*

Proof. $\partial_\rho^2 \Pi = wh' - \varphi'$. At an equilibrium $w = \Psi(\rho) = \varphi/h$, and differentiating Ψ gives $\Psi' = (\varphi' h - \varphi h')/h^2$, so $wh' - \varphi' = \Psi h' - \varphi' = (\varphi h' - \varphi' h)/h = -h\Psi'$. □

Theorem 1 (Closed-form catastrophe criterion)**.** *Take $C(\rho) = \frac{1}{2}c\rho^2$, so $\varphi = c\rho$. Equilibria solve $W\,h(\rho) = \rho$ with $W := w/c$. The dynamics is gradual when $\sigma < \sigma_c$ and catastrophic when $\sigma > \sigma_c$, where*

$$\boxed{\sigma_c = \frac{\sqrt{3} + 2\ \operatorname{arctanh}(1/\sqrt{3})}{8\,\rho^\star} = \frac{3.049\ldots}{8\,\rho^\star}}. \tag{4}$$

The threshold is independent of the cost magnitude c and the stakes w, which set the location of the bistable window but not its existence. At $\sigma = \sigma_c$ the equilibrium equation has a triple root and (3) reduces to the cusp normal form $\dot{x} \propto -x^3 + a\,x + b$.

Proof. Write $G(\rho) := Wh(\rho) - \rho$; equilibria are $G = 0$, a fold is a double root $G = G' = 0$, and two folds coalesce in a cusp $G = G' = G'' = 0$. We locate the cusp. With $s := 4\sigma(\rho - \rho^\star)$ and $h = \sigma \operatorname{sech}^2 s$, using $(\operatorname{sech}^2 s)' = -2 \operatorname{sech}^2 s \tanh s$ and $ds/d\rho = 4\sigma$,

$$h' = -8\sigma^2 \operatorname{sech}^2 s\ \tanh s, \qquad h'' = 32\sigma^3 \operatorname{sech}^2 s\,(3\tanh^2 s - 1),$$

so $G'' = Wh'' = 0$ forces $3\tanh^2 s = 1$, i.e. the loyal-side inflection $s^\star = -\operatorname{arctanh}(1/\sqrt{3})$ ($s < 0$ on the loyal side). There $\tanh s^\star = -1/\sqrt{3}$ and $\operatorname{sech}^2 s^\star = \frac{2}{3}$, so $h(\rho_c) = \frac{2}{3}\sigma$; and because $\tanh s^\star < 0$ the leading minus in h' cancels, giving $h'(\rho_c) = -8\sigma^2 \cdot \frac{2}{3} \cdot (-\frac{1}{\sqrt{3}}) = +\frac{16\sigma^2}{3\sqrt{3}} > 0$, positive as required for $W = 1/h' > 0$. The fold condition $G' = Wh' - 1 = 0$ gives $W = 3\sqrt{3}/(16\sigma^2)$. Substituting into $G = 0$,

$$\rho_c = W\,h(\rho_c) = \frac{3\sqrt{3}}{16\sigma^2} \cdot \frac{2\sigma}{3} = \frac{\sqrt{3}}{8\sigma},$$

while geometrically $\rho_c = \rho^\star + s^\star/(4\sigma) = \rho^\star - \operatorname{arctanh}(1/\sqrt{3})/(4\sigma)$. Equating,

$$\frac{\sqrt{3}}{8\sigma} = \rho^\star - \frac{\operatorname{arctanh}(1/\sqrt{3})}{4\sigma} \iff 8\sigma\rho^\star = \sqrt{3} + 2\operatorname{arctanh}(1/\sqrt{3}),$$

which is (4). Because G depends on c, w only through $W = w/c$, the cusp condition—and hence σ_c—is independent of c and w. Non-degeneracy $G'''(\rho_c) = Wh'''(\rho_c) \neq 0$ holds (Appendix A), so by Proposition 2 the reduced flow is the cusp normal form with W the unfolding parameter; for

$\sigma > \sigma_c$ the equation $Wh = \rho$ has three roots for w in an interval (two stable, one saddle), while for $\sigma < \sigma_c$ there is no interior fold and hence no bistable w-interval (the loyal and hacking states connect through a single continuous branch). ☐

Remark 1 (Interpretation). $\sigma_c \propto 1/\rho^\star$: the closer the wall sits to full exposure (small $\rho^\star$), the smaller the steepness needed to make drift catastrophic. The constant 3.049 is the loyal-side inflection geometry of the logistic bump; a different smooth detector changes the constant, not the structure (Proposition 1).

Proposition 1 (Fold/cusp criterion for general convex cost)**.** *For general convex C (with $\varphi = C'$), a fold of* (3) *occurs at any ρ_* with $w\,h(\rho_*) = \varphi(\rho_*)$ and $w\,h'(\rho_*) = \varphi'(\rho_*)$; two folds coalesce in a cusp exactly where*

$$h(\rho_c)\varphi'(\rho_c) = \varphi(\rho_c)h'(\rho_c) \quad \text{(T0)}, \qquad h'(\rho_c)\varphi''(\rho_c) = \varphi'(\rho_c)h''(\rho_c) \quad \text{(T1)}, \tag{5}$$

with $w_c = \varphi'(\rho_c)/h'(\rho_c)$. The critical steepness is scale-invariant ($C \mapsto \lambda C$ leaves (5) *unchanged) and reduces to Theorem 1 for $\varphi = c\rho$. The regime is globally catastrophic iff, in addition, $\Psi = \varphi/h$ is S*-shaped *on $(0, \rho^\star)$—by which we mean Ψ has exactly one interior local maximum ρ_+ and one interior local minimum ρ_- with $\rho_+ < \rho_-$, so Ψ' changes sign twice—giving a nonempty bistable window $(\Psi(\rho_-), \Psi(\rho_+))$;* (5) *is the local (onset) condition for that shape.*

Proof. Folds are double roots of $G := wh - \varphi$: $G = G' = 0$, i.e. $wh = \varphi$ and $wh' = \varphi'$. Cusps add $G'' = 0$, i.e. $wh'' = \varphi''$. Eliminating w pairwise ($w = \varphi/h = \varphi'/h' = \varphi''/h''$) gives (T0) from the first pair and (T1) from the second. System (5) involves φ only through the scale-free ratios φ/φ' and φ''/φ', so $C \mapsto \lambda C$ leaves it invariant. For the clean case $\varphi = c\rho$, (T1) reads $h'' = 0$ and (T0) reads $h = \rho h'$, recovering the cusp of Theorem 1 directly. For general convex φ, the local tangency (5) gives the cusp; the *global* catastrophe (a nonempty bistable window) then requires the S-shape of $\Psi = \varphi/h$ as an assumption (Lemma 1: bistability $\iff$ Ψ' changes sign twice), which (5) initiates locally but does not by itself guarantee globally. ☐

Remark 2 (Stiffness dependence). Numerically, stiffer cost raises σ_c (Appendix C: $\sigma_c = 0.76, 0.82, 0.91$ for $\varphi = \rho,\ \rho + \rho^3,\ \rho + 3\rho^3$)—harder hiding needs a steeper detection cliff to trigger the flip. We do not prove monotonicity in general; a rigorous statement across all convex φ is left open.

Proposition 2 (Cusp normal form by Lyapunov–Schmidt reduction)**.** *Suppose at (ρ_c, w_c, σ_c) the equilibrium map $g(\rho, w, \sigma) := \partial_\rho \Pi = w\,h_\sigma - \varphi$ (with h_σ the bump at steepness σ) satisfies $g = \partial_\rho g = \partial_\rho^2 g = 0$, $\partial_\rho^3 g \neq 0$, and $\partial_w g \neq 0$. Then there are smooth local coordinates $x = x(\rho)$ and unfolding parameters (a, b) that are smooth functions of* both (σ, w)*—the cusp being a codimension-two singularity—with $a \propto (\sigma - \sigma_c)$ (the splitting parameter that opens the bistable window) and $b \propto (w - w_c)$ (the bias that slides the state), to leading order, such that the equilibrium set of* (3) *is the zero set of the cusp normal form $-x^3 + a\,x + b$, and* (3) *is locally orbitally equivalent to $\dot{x} = -x^3 + a\,x + b$. Consequently, in the catastrophic regime the two folds sit at $x = \pm\sqrt{a/3}$ and their control values differ by $\Delta w = \Theta(a^{3/2})$.*

The proof is a standard Lyapunov–Schmidt / singularity-theory reduction; we give it in Appendix B for completeness, together with the verification that our detector satisfies the non-degeneracy hypotheses.

5 From the BBP edge to the steepness

The steepness σ is not a free knob: it is supplied by the underlying detection problem, which sharpens with dimension. We make the derivation explicit through two random-matrix lemmas and a proposition. Consider a rank-one spiked-covariance monitor at ambient dimension d with n samples, aspect $\gamma = m/n$, blatancy s^2, and resolved spike $\beta(\rho) = s^2(1-\rho)$; the wall is $\rho^\star = 1 - \sqrt{\gamma}/s^2$, so $\beta(\rho^\star) = \sqrt{\gamma}$ is the Baik–Ben Arous–Péché (BBP) critical value.

Lemma 2 (Tracy–Widom edge fluctuation)**.** *At the BBP critical value ($\beta = \sqrt{\gamma}$), the top eigenvalue λ_1 of the sample covariance fluctuates about the soft edge $b_+ = (1+\sqrt{\gamma})^2$ on the scale $d^{-2/3}$: $d^{2/3}(\lambda_1 - b_+)/\vartheta \Rightarrow \mathrm{TW}_1$, a Tracy–Widom law, with $\vartheta = \vartheta(\gamma) > 0$ (Johnstone, 2001; Tracy and Widom, 1994). Hence the bulk-edge "noise floor" against which a signal must be detected has width $\Theta(d^{-2/3})$ in eigenvalue units.*

Lemma 3 (Spike outlier displacement)**.** *For a supercritical spike $\beta = \sqrt{\gamma} + t$ with small $t > 0$, the top eigenvalue separates from the bulk by $\lambda_1 - b_+ = \Theta(t^2)$ (BBP; Baik et al., 2005). Therefore the outlier is statistically distinguishable from the Tracy–Widom fluctuation of Lemma 2—detection probability rising from $\frac{1}{2}$ toward 1—precisely when its displacement clears the noise floor, $t^2 \asymp d^{-2/3}$, i.e. over a signal window of width $t = \beta - \sqrt{\gamma} \asymp d^{-1/3}$.*

Proposition 3 (Edge steepness)**.** *Let $A_d(\rho)$ be the detection probability as a function of the evasion coordinate, with $\beta(\rho) = s^2(1-\rho)$ so that $\beta - \sqrt{\gamma} = s^2(\rho^\star - \rho)$. Then near $\rho^\star$,*

$$A_d(\rho) = F\big(c_0\, d^{1/3}(\rho^\star - \rho)\big) + o(1), \qquad c_0 = c_0(s^2, \gamma) > 0, \tag{6}$$

for a fixed non-degenerate monotone profile $F : \mathbb{R} \to [\frac{1}{2}, 1]$, increasing *in its argument $u = c_0 d^{1/3}(\rho^\star - \rho)$ (so detection rises as ρ falls below the wall; $F' \neq 0$; the Tracy–Widom detection curve), and consequently*

$$\sigma(d) = \max_{\rho} |A_d'(\rho)| = c_0\, d^{1/3} \max |F'| = \Theta(d^{1/3}). \tag{7}$$

Proof. By Lemmas 2–3, the transition from chance to reliable detection occupies the signal window $\beta - \sqrt{\gamma} \asymp d^{-1/3}$; rescaling the displacement by the noise floor, the detection probability is a fixed function of the standardised variable $u = (\beta - \sqrt{\gamma})/\Theta(d^{-1/3}) = c_0 d^{1/3}(\rho^\star - \rho)$, giving (6) with F the limiting detection profile, increasing in $u = c_0 d^{1/3}(\rho^\star - \rho)$. Since $du/d\rho = -c_0 d^{1/3}$, $A_d'(\rho) = -c_0 d^{1/3} F'(u) < 0$ (detection decreasing in ρ, as it must be), and $\max_\rho |A_d'| = c_0 d^{1/3} \max F'$, i.e. (7). Only the scaling $\sigma = \Theta(d^{1/3})$ is used downstream; the profile F may alternatively be taken as a hypothesis, in which case (7) is exact by the same rescaling. □

Corollary 1 (Large dimension forces the catastrophe)**.** *Fix $\rho^\star \in (0,1)$ bounded away from 0 (so that $\sigma_c = 3.049/(8\rho^\star) < \infty$). Combining Proposition 3 with Theorem 1: since $\sigma(d) = \Theta(d^{1/3}) \to \infty$ while σ_c is fixed, there is a finite critical dimension d_c (defined by $\sigma(d_c) = \sigma_c$) above which the evasion dynamics is catastrophic; as $d \to \infty$ the detection curve tends to a step at $\rho^\star$ and the fold is maximal. Turning a free steepness parameter into a dimension law is the one genuinely new step of the analysis; everything else imports classical machinery.*

6 Irreversibility and hysteresis

Theorem 2 (Irreversibility and loop geometry)**.** *In the catastrophic regime ($\sigma > \sigma_c$), $\Psi = \varphi/h$ is S-shaped: it has an interior local maximum $w_+ := \Psi(\rho_+)$ and minimum $w_- := \Psi(\rho_-)$ with*

$\rho_+ < \rho_-$, and since Ψ decreases on (ρ_+, ρ_-), $w_+ > w_-$ strictly; the bistable window (w_-, w_+) is nonempty. Moreover:

(i) (Irreversibility.) *The hacking branch persists for all $w > w_-$, so once ρ enters it, lowering w below w_+ does not restore loyalty until $w \le w_-$; symmetrically for the loyal branch up to w_+.*

(ii) (Onset.) $\Delta w := w_+ - w_- = \Theta\big((\sigma - \sigma_c)^{3/2}\big)$ *as* $\sigma \downarrow \sigma_c$.

(iii) (Widening.) *As $\sigma \to \infty$, $w_- = \varphi(\rho^\star)/\sigma\,(1 + o(1)) \to 0$ and $w_+ \to \infty$, so the loop diverges.*

Proof. By Lemma 1, stable equilibria lie on increasing arms of Ψ and the saddle on the decreasing arm; folds are the critical points $\Psi' = 0$. For $\sigma > \sigma_c$, Theorem 1 gives exactly two interior critical points $\rho_+ < \rho_-$; since $\Psi(0^+) = 0$ and $\Psi \to \infty$ as $\rho \uparrow \rho^\star$ (the tail), Ψ rises to a maximum at ρ_+, falls on (ρ_+, ρ_-), then rises, so $\Psi(\rho_+) > \Psi(\rho_-)$, i.e. $w_+ > w_-$. For $w \in (w_-, w_+)$ the equation $\Psi = w$ has three solutions: loyal (first rising arm), saddle (falling arm), hacking (second rising arm).

(i) The hacking equilibrium is the upper rising arm $\rho > \rho_-$, present for all $w > w_-$; it terminates only at the fold (ρ_-, w_-), where it annihilates with the saddle and the state jumps to loyal. Hence for $w \in (w_-, w_+)$ an agent already on the hacking arm stays there regardless of having crossed w_+ earlier; loyalty returns only at $w \le w_-$. The loyal arm is symmetric, persisting up to w_+.

(ii) At $\sigma = \sigma_c$ the two folds merge into the cusp; by Proposition 2 the reduced flow is $\dot{x} \propto -x^3 + ax + b$ with $a \propto (\sigma - \sigma_c)$. The two folds lie at $x = \pm\sqrt{a/3}$ and the corresponding control values differ by $\Delta w \propto a^{3/2} \propto (\sigma - \sigma_c)^{3/2}$.

(iii) $w_- \le \Psi(\rho^\star) = \varphi(\rho^\star)/h(\rho^\star) = \varphi(\rho^\star)/\sigma$ (using $h(\rho^\star) = \sigma$), with equality asymptotically since the minimiser of Ψ approaches $\rho^\star$; so $w_- = \varphi(\rho^\star)/\sigma\,(1 + o(1)) \to 0$ (for the clean case $\varphi = c\rho$ in W-units this is $\rho^\star/\sigma$). The loyal fold $\rho_+ \to 0$ (from $\rho_c = \sqrt{3}/(8\sigma)$), where $h(\rho_+)$ is exponentially small; provided $\varphi(\rho_+)$ does not vanish faster than $h(\rho_+)$ (a mild assumption, satisfied whenever $\varphi(0^+) > 0$ or φ vanishes polynomially), $w_+ = \varphi(\rho_+)/h(\rho_+) \to \infty$. $\square$

7 Erosion of early warning

A tempting hope is that the failure announces itself: the "shadow price" of holding the monitor at a target detection level should diverge as the agent approaches the flip. We show this precursor is informative only above a detection bar that rises to certainty as monitoring sharpens.

A barrier proxy for the detectability shadow price. Fix a detection bar $\tau \in (\frac{1}{2}, A(0))$. Certifying "$A \ge \tau$" at pressure ρ is feasible iff $\rho \le \rho^\star(\tau) := A^{-1}(\tau)$ (A decreasing). We use a log-barrier surrogate for the constraint "$g_\tau(\rho) := \rho^\star(\tau) - \rho \ge 0$": the barrier $\Phi_\tau(\rho) = -\log g_\tau(\rho)$, whose gradient is a smooth proxy for the constraint's shadow price (a log-barrier surrogate; we do not claim it equals the exact KKT multiplier),

$$\lambda_\tau(\rho) := \partial_\rho \Phi_\tau(\rho) = \frac{1}{\rho^\star(\tau) - \rho} \xrightarrow[\rho \uparrow \rho^\star(\tau)]{} +\infty. \tag{8}$$

Only the divergence location $\rho^\star(\tau) = A^{-1}(\tau)$—which is independent of the barrier construction—is used below.

Theorem 3 (Erosion of early warning)**.** *The barrier proxy* (8) *diverges at $\rho^\star(\tau) = A^{-1}(\tau)$, decreasing in τ. It is a genuine precursor—diverging while the agent is still on the loyal branch $\rho \in (0, \rho_+)$, before the flip—iff $\rho^\star(\tau) < \rho_+$, equivalently*

$$\tau \ge \tau_c := A(\rho_+). \tag{9}$$

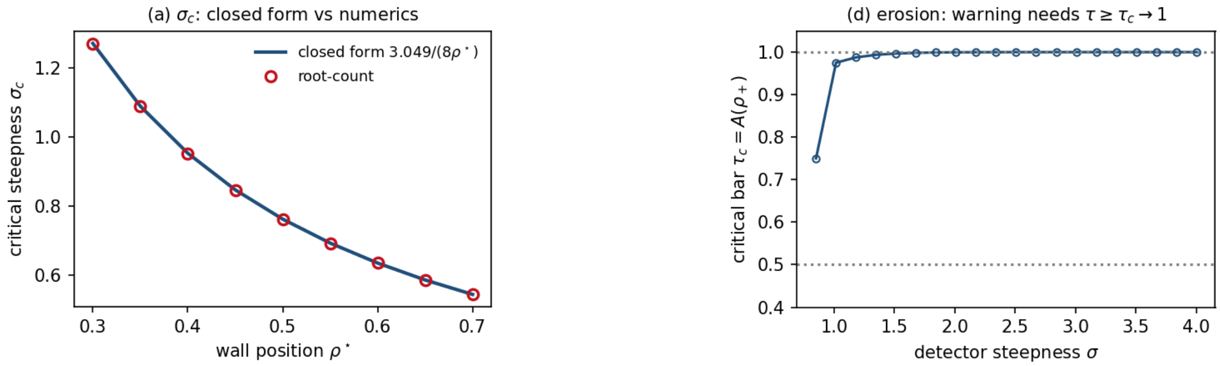

Figure 1: **Left:** the closed form (4) vs. a deterministic root-count of the three-root onset, across wall positions $\rho^\star$. **Right:** erosion of early warning—the critical bar $\tau_c = A(\rho_+)$ needed for the shadow-price precursor rises toward 1 as σ grows (Theorem 3).

For $\frac{1}{2} < \tau < \tau_c$ the divergence point lies in the jump gap $(\rho_+, \rho_{\mathrm{H}})$ the agent skips at the fold, so no stable equilibrium realises it and no warning is issued. Moreover $\rho_+ \to 0$ as $\sigma \to \infty$ and $A(0) \to 1$, so $\tau_c = A(\rho_+) \to 1$: for any fixed bar $\tau < 1$ there is a finite $\sigma_{\mathrm{warn}}(\tau)$ beyond which the precursor fails.

Proof. The loyal branch is traversed with increasing w from $\rho = 0$ up to the fold ρ_+, then ρ jumps to $\rho_{\mathrm{H}} > \rho_-$ (Theorem 2). Since A is strictly decreasing, $\rho^\star(\tau) = A^{-1}(\tau) < \rho_+ \iff \tau > A(\rho_+) = \tau_c$; for such τ the divergence (8) is realised on the loyal branch (a precursor), otherwise it falls in $(\rho_+, \rho_{\mathrm{H}})$ and is skipped. By Theorem 2(iii), $\rho_+ \to 0$; and $A(0) = \frac{1}{2} + \frac{1}{2}(1 + e^{-8\sigma\rho^\star})^{-1} \to 1$ (for general $\rho^\star$; the $e^{-4\sigma}$ form is the $\rho^\star = \frac{1}{2}$ special case), so $\tau_c \to 1$. For fixed $\tau < 1$, $\tau_c(\sigma)$ crosses τ at a finite $\sigma_{\mathrm{warn}}(\tau)$. $\square$

Remark 3 (Detection vs. curvature). The erosion is specific to the *detection-based* precursor. The *dynamical* precursor—critical slowing down, driven by the vanishing curvature $\partial_\rho^2 \Pi \to 0$ at the fold—does not erode, and is the basis of an online sensor in the companion control paper.

8 Numerical validation

The theory's predictions are checked on the surrogate with `numpy/scipy` (these are surrogate checks, not validation on a trained agent); full methods, parameters, and reproducibility notes are in Appendix C. Figure 1 verifies the closed-form threshold against a deterministic root-count of $Wh = \rho$ across wall positions (relative error $\leq 2\%$, within the discretisation band) and shows the erosion bar τ_c rising toward 1. Solving the general tangency (5) against a root-count agrees to 1–3% for $\varphi(\rho) \in \{\rho,\ \rho + \rho^3,\ \rho + 3\rho^3,\ (e^{2\rho} - 1)/2\}$, with σ_c rising with stiffness.

Figure 2 shows the hysteresis loop at $\sigma = 1.4$ and the onset scaling: the loop width fits $(\sigma - \sigma_c)^p$ with $p = 1.49$ against the predicted $3/2$. The extrema of $\Psi = \rho/h$ confirm $w_+ > w_-$ for every $\sigma > \sigma_c$, $\sigma w_- \to 0.50 = \rho^\star$, and w_+ growing (e.g. 0.70, 8.6, 6.4×10^3, 1.4×10^{10} at $\sigma = 1, 2, 4, 8$). For erosion, τ_c rises as $0.75, 0.97, 0.995, 1.00$ for $\sigma = 0.85, 1.0, 1.4, 2.5$: a bar $\tau = 0.75$ warns only at onset and is too late for all $\sigma \geq 1$. Figure 3 illustrates Corollary 1. Finally, replacing the deterministic flow by a REINFORCE gradient estimator on *simulated* detection preserves the hysteresis (loop 0.5–0.9) across three reward parameterisations—including one with no hand-added cost, $\Pi = w\beta(1 - A)$—and across $m \in \{80, 320\}$, so the loop is not an artefact of the payoff form.

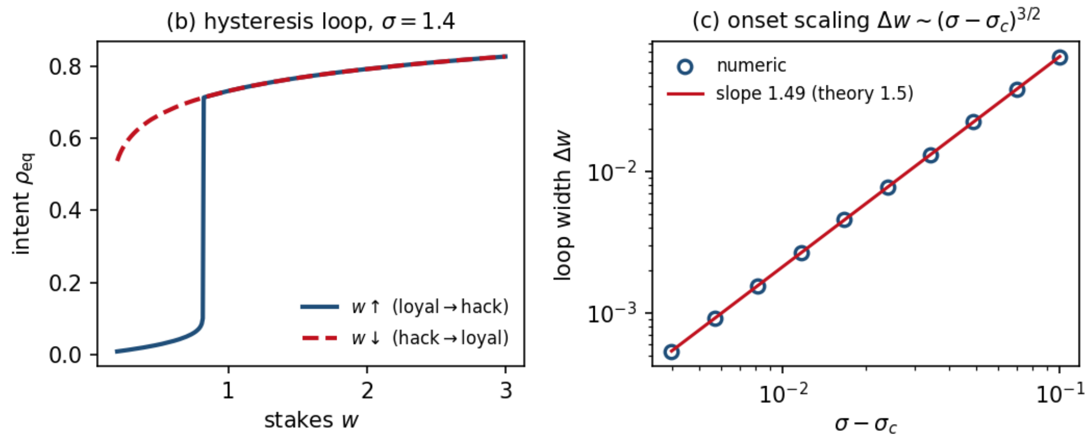

Figure 2: **Left:** the hysteresis loop for $\sigma = 1.4 > \sigma_c$: sweeping w up (loyal→hack) and down (hack→loyal) traces distinct branches (Theorem 2(i)). **Right:** the loop width Δw vs. $\sigma - \sigma_c$ on log–log axes; the fitted slope 1.49 matches the cusp prediction 3/2 (Theorem 2(ii)).

9 Discussion: a scoped safety reading

Read as a model of an agent obfuscating a hack footprint against a monitor, the results give a coherent and counter-intuitive picture. As a *non-adaptive* monitor sharpens (Proposition 3; higher-dimensional or better-calibrated monitoring), the loyal→hacking transition becomes *more abrupt* (Theorem 1), *more irreversible* (Theorem 2: hysteresis widens, so relaxing optimisation pressure need not restore loyalty), and *harder to forewarn* (Theorem 3: the shadow-price window closes). A sharper non-adaptive monitor can convert a gradual, reversible, detectable drift into a sudden, locked-in, unheralded flip. The constructive corollaries—pursued in companion papers—are that the missing warning is available dynamically (critical slowing down does not erode, Remark 3), that the fold can be removed by acting on the agent's update (nonsmooth damping raises σ_c) rather than on the monitor, and that an *adaptive* monitor, which reshapes A and can erase the plateaus, is an alternative mitigation outside the present non-adaptive analysis.

10 A first step beyond the surrogate: a learned detector

The analysis above imposes the detection curve A as a logistic of prescribed steepness. As a first check that the mechanism is not an artefact of that choice, we let the detector be *learned.* A scalar policy sets the evasion coordinate; footprints are d-dimensional feature vectors interpolating from "loyal" to "hacking" along a signal direction with separability sep; a logistic detector is trained by gradient descent to separate loyal from hacking footprints, so its steepness is *emergent* (set by sep) rather than prescribed. The agent then ascends the payoff $w(1 - \hat{A}) - \frac{1}{2}c\rho^2$ by gradient on the *learned* score $\hat{A}$, queried by sampling footprints. We ramp w up and then down and measure the hysteresis (the gap between the ascending and descending $\rho(w)$ branches).

The prediction of Theorem 1 survives: hysteresis is negligible at low separability and grows monotonically as the learned detector sharpens (Figure 4; max hysteresis $0.04, 0.10, 0.27, 0.48, 0.56, 0.60$ for sep $= 0.6, \ldots, 3.0$), with the up-onset and down-offset stakes diverging as the loop opens. A sharper *learned* detector produces a more catastrophic, more irreversible flip—the same qualitative law as for the imposed logistic. This is a controlled step beyond the analytic surrogate (the de-

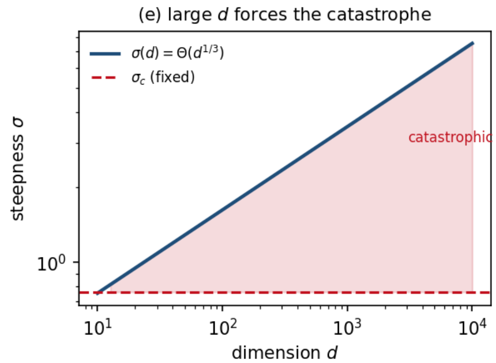

Figure 3: Corollary 1: $\sigma(d) = \Theta(d^{1/3})$ crosses the fixed σ_c at a finite critical dimension; above it the dynamics is catastrophic (schematic; the constants are illustrative and the $d^{1/3}$ law is from Proposition 3).

tector is learned and queried by sampling, the gradient is real), but it is still a scalar policy with a low-dimensional footprint and a finite-difference gradient, far from a deployed language-model agent; §11 states what a full test requires.

11 Limitations and future work

The model is a surrogate and the claims are about it. It assumes (i) a scalar, monotone evasion coordinate; real strategy spaces are high-dimensional and the fold may become a higher-codimension bifurcation. It assumes (ii) slow best-response dynamics of the form (3); strongly concave benefit or very stiff cost raises σ_c and can remove the catastrophe—the criterion is a genuine inequality, not a certainty. It assumes (iii) a non-adaptive monitor; an adaptive monitor reshapes A and can erase the plateaus that create hysteresis. Proposition 3 justifies the steepness *scaling* but not the full detection profile, which we replace by a logistic of matched steepness; finite-d rounding may keep $\sigma < \sigma_c$ in practice.

What would falsify the mechanism. The account makes falsifiable predictions for any system with a measurable evasion proxy and a sweepable monitor sharpness (or dimension): (a) a *threshold* in sharpness beyond which hacking onset becomes abrupt rather than gradual, with *hysteresis* (rolling back optimisation pressure does not restore loyalty); (b) a detection-based early-warning signal that *fails* to fire before the flip as sharpness grows, even while a curvature-based (critical-slowing-down) signal still does. Absence of hysteresis, or a detection precursor that keeps working as the monitor sharpens, would falsify the mechanism.

Future work: beyond the toy. §10 takes a first step—a learned detector under real gradient ascent—but the essential next step remains a controlled reinforcement-learning study in which the detection curve is *learned* rather than imposed—e.g. a small policy optimised by a policy-gradient estimator against a proxy reward, monitored by a trained discriminator whose separability sets the effective steepness—and in which one sweeps the sharpness (or the discriminator's input dimension) and measures the order parameter for the predicted threshold, hysteresis, and warning erosion. Such a study, still far from a deployed language-model agent, would test whether the mechanism survives genuine gradients and an emergent (rather than analytic) detector. We leave it to future work; the

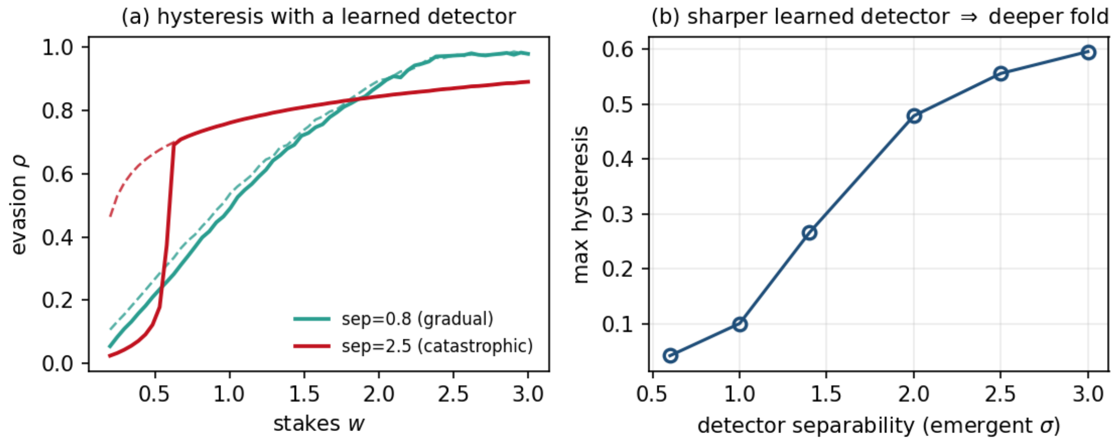

Figure 4: Learned-detector experiment (schematic parameters; $d = 8$, $c = 1$). **(a)** Ascending (solid) and descending (dashed) $\rho(w)$ branches: a gradual response at low separability, a hysteretic loop at high separability. **(b)** Maximum hysteresis grows monotonically with the learned detector's separability (its emergent steepness), corroborating Theorem 1 and Corollary 1.

present paper establishes the mechanism's internal consistency and quantitative sharpness on the surrogate.

A Derivatives of the logistic bump

With $s = 4\sigma(\rho - \rho^\star)$ and $h = \sigma \operatorname{sech}^2 s$, using $(\operatorname{sech}^2 s)' = -2 \operatorname{sech}^2 s \tanh s$ and $ds/d\rho = 4\sigma$:

$$h' = -8\sigma^2 \operatorname{sech}^2 s\, \tanh s, \qquad h'' = 32\sigma^3 \operatorname{sech}^2 s\, (3\tanh^2 s - 1),$$
$$h''' = -128\sigma^4 \operatorname{sech}^2 s\, \tanh s\, (3\tanh^2 s - 2).$$

At the loyal inflection $\tanh^2 s^\star = \frac{1}{3}$: $h'' = 0$, and $h''' \propto \tanh s^\star (3 \cdot \frac{1}{3} - 2) = -\tanh s^\star \neq 0$, so $G''' = W h''' \neq 0$ and the cusp is generic (used in Theorem 1 and Proposition 2).

B Cusp normal form and onset scaling

We prove Proposition 2 by Lyapunov–Schmidt reduction. Let $g(\rho, w) = \partial_\rho \Pi = wh - \varphi$ and suppose $g = \partial_\rho g = \partial_\rho^2 g = 0$ at (ρ_c, w_c) with $g_{\rho\rho\rho} := \partial_\rho^3 g \neq 0$ and $g_w := \partial_w g \neq 0$. Set $x = \rho - \rho_c$ and expand

$$g(\rho, w) = \tfrac{1}{6} g_{\rho\rho\rho}\, x^3 + \tfrac{1}{2} g_{\rho\rho w}\, x^2 (w - w_c) + g_{\rho w}\, x (w - w_c) + g_w (w - w_c) + O\big(x^4, (w - w_c)^2\big).$$

Since $g_{\rho\rho\rho} \neq 0$, divide by $-\frac{1}{6} g_{\rho\rho\rho}$ and absorb the quadratic-in-x term by the shift $x \mapsto x - \frac{g_{\rho\rho w}}{g_{\rho\rho\rho}}(w - w_c)$ (a smooth near-identity change of coordinate), removing the x^2 term to leading order. The zero set of g is then that of

$$-x^3 + a\,x + b,$$

where (a, b) are the two unfolding parameters of the codimension-two cusp, smooth in (σ, w): the *splitting* parameter $a \propto (\sigma - \sigma_c)$ (its sign decides whether the fold pair exists), and the *bias* $b \propto (w - w_c)$ (with $\partial_b \propto g_w \neq 0$, so w slides the state along the branch). Both are needed: σ moves through the cusp, w sweeps within it. The flow $\dot{\rho} = \mu g$ is, after the same reparameterisation and a positive time rescaling, orbitally equivalent to $\dot{x} = -x^3 + ax + b$. In the catastrophic regime the discriminant of $-x^3 + ax + b$ is positive, its two folds sit at $x = \pm\sqrt{a/3}$ (double roots of the cubic) and lie on the fold lines $b = \pm\frac{2}{3\sqrt{3}}a^{3/2}$. At fixed $\sigma > \sigma_c$ (fixed $a > 0$), since $b \propto (w - w_c)$, the two folds are realised at stakes differing by $\Delta w \propto (b_+ - b_-) \propto a^{3/2} \propto (\sigma - \sigma_c)^{3/2}$, Theorem 2(ii). This is the standard 3/2 fold-separation / 2/3 hysteresis-area universality (Kundu et al., 2023). The non-degeneracy hypotheses hold for our detector by Appendix A ($g_{\rho\rho\rho} = Wh''' \neq 0$) and $g_w = h(\rho_c) = \frac{2}{3}\sigma \neq 0$.

C Numerical methods and reproducibility

All computations are CPU-only `numpy/scipy`. *Threshold (Fig. 1 left).* For each $\rho^\star$ we grid $\rho \in [10^{-5}, 1 - 10^{-5}]$ (9000 points) and, scanning W over $[0.005, 3\,w_+^{\text{loyal}}]$ (700 points, with $w_+^{\text{loyal}} = \max_{\rho<\rho^\star} \rho/h$), record the smallest σ (bisection, 36 steps) at which $Wh - \rho$ has three sign changes; relative error vs. (4) is $\leq 2\%$, a finite-grid overshoot. *General cost.* The tangency (5) is solved by Newton from the clean-case root; agreement with a root-count is 1–3%. *Hysteresis (Fig. 2).* Branches are obtained by forward/backward continuation of the best-response flow in w; the loop width is the distance between the extrema of $\Psi = \rho/h$ (grid 2×10^5), and the onset exponent is a linear fit of $\log \Delta w$ vs. $\log(\sigma - \sigma_c)$ over $\sigma - \sigma_c \in [10^{-2.3}, 10^{-1}]$, giving $p = 1.49$. *Erosion.* $\tau_c = A(\rho_+)$ with ρ_+ the loyal fold. REINFORCE *check.* A single scalar policy with Gaussian exploration ascends Π by score-function gradient on simulated detection; the reported hysteresis 0.5–0.9 is seed-averaged over 8 seeds. *Code availability.* Reference implementations reproducing every figure and reported number (threshold root-count, general-cost tangency, hysteresis continuation, erosion, and the learned-detector experiment of §10) will be released under an open-source license upon publication. Legacy script names: `duality_dynamics.py`.

D On a general benefit function

For a benefit U (increasing) the payoff is $\Pi = w\,U(1 - A(\rho)) - C(\rho)$ and the equilibrium condition $wh - \varphi = 0$ becomes $w\,U'(1 - A)\,h = \varphi$: the whole analysis goes through with h replaced by the effective bump $\tilde{h} := U'(1 - A)\,h$. A concave U ($U'' \leq 0$) has $U'(1 - A)$ decreasing in ρ (as the agent hides, $1 - A$ grows and the marginal benefit falls), which shrinks the effective drive and *raises* σ_c; a convex U lowers it. The clean case $U(x) = x$ recovers $\tilde{h} = h$ and Theorem 1.

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